



HACK2IGNITE 2026

TEAM NAME:- BYTE BATTLE

PROBLEM STATEMENT ID:- AI-03

INSTITUTION NAME:- G. H. RAISONI COLLEGE OF ENGINEERING AND
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TEAM MEMBERS:-

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Proposed Solution

SATYASET AI is an AI-powered, multilingual verification platform that checks text, images, videos and URLs using trusted sources and evidence.

How It Addresses the Problem

- Extracts key claims from shared content.
- Checks facts and context against reliable sources.
- Detects false claims, misleading context and manipulated media.
- Provides an explainable verdict with supporting evidence.
- Gives corrective information in Indian languages.

Innovation & Uniqueness

- Multilingual
- Context-Aware
- Sector-Aware
- Evidence-Based

USP

“Not just Fake or Real — SATYASET AI shows the evidence behind the verdict.”



TECHNICAL APPROACH

TECHNOLOGY • METHODOLOGY • IMPLEMENTATION

03 TECHNOLOGY STACK

TECHNOLOGY | METHODOLOGY | IMPLEMENTATION



1 Frontend

React + Tailwind CSS

User-friendly
and responsive
interface



2 Backend

FastAPI / Node.js + PostgreSQL

Secure and
scalable backend



3 AI / ML Models

LLM + MuRIL / Indic Models
+ Sentence Transformers

Multilingual
and context-aware
analysis



4 Verification Engine

RAG + FAISS/ChromaDB
+ Google Fact Check Tools API

Trusted source
retrieval and
fact-checking



5 Media Analysis Tools

OCR + Whisper + OpenCV + CLIP

Process and
analyze images,
videos and audio



6 Deployment & Infrastructure

Docker + Cloud (AWS/GCP)

Scalable and
reliable deployment



7 Monitoring & Updates

Continuous model improvement
and source updates

Ensure accuracy
and trust over time



Key Principle

“AI + Trusted Sources
for a more informed India.”

03 METHODOLOGY & PROCESS

FROM INPUT TO TRUSTED INFORMATION



1 User Input

Text | Image | Video | URL

Supports
multiple formats



2 Pre-processing

OCR | Speech-to-Text | Language Detection

Convert and
prepare data



3 Claim Analysis

Extract key claims | Classify | Understand context

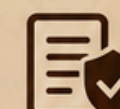
Identify what
needs to be verified



4 Evidence Search

RAG | Trusted Sources | Fact-Check Tools

Retrieve relevant
and reliable
information



5 Verification

Check facts | Context | Media authenticity

Detect false or
misleading content



6 Evidence Fusion

Compare multiple sources | Analyse | Score

Ensure accurate
and balanced
verdict



7 Final Output

Verdict | Sources | Correct Information

Clear explanation
with evidence
(in Indian languages)



Key Principle

“AI assists verification;
evidence supports the decision.”

03 IMPLEMENTATION

FROM VISION TO REAL IMPACT



1 Requirement Analysis

Study user needs, define features
and set objectives.

Understand the
problem and
user requirements



2 System Design

Design architecture, select
technologies and define modules.

Plan a scalable,
secure and
modular solution



3 Development

Build and integrate frontend,
backend, AI/ML and verification modules.

Agile development
with continuous
integration



4 Testing & Validation

Test for accuracy, performance,
security and real-world scenarios.

Ensure reliability
and robustness



5 Deployment

Deploy on cloud (AWS/GCP) with
monitoring and scalable infrastructure.

Make the platform
accessible to users



6 User Feedback & Iteration

Collect user feedback, improve models
and add new features.

Continuous
improvement for
greater impact



Our Goal

“A reliable, accessible and impactful
fact-verification platform for **every Indian.**”

04

FEASIBILITY & RISK MANAGEMENT

ANALYSIS | CHALLENGES | STRATEGIES | WAY FORWARD



FEASIBILITY OF THE IDEA

Practical | Scalable | Impactful



Software-based solution

No special hardware required



Modular architecture

Easy to develop and upgrade



Existing AI models & APIs

Faster MVP development



Trusted-source registry

Evidence-based verification



MVP-first approach

Start small, scale gradually



POTENTIAL CHALLENGES & RISKS

Identifying risks early for a stronger solution



AI Hallucination

May generate incorrect or misleading information



Missing or Limited Evidence

Reliable sources may not always be available



Language Ambiguity

Different meanings in regional languages



Deepfake & Manipulated Media

Fake images/videos can be hard to detect



Source Bias

Sources may be biased or politically influenced



Privacy & Data Security

User data and uploaded content need protection



STRATEGIES FOR OVERCOMING THESE CHALLENGES

Turning challenges into opportunities



RAG + Citations

Use trusted sources and always provide references



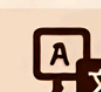
Unverified State

Show “insufficient evidence” instead of wrong answers



Cross-Source Checks

Validate information from multiple reliable sources



Multilingual Models

Use and fine-tune models for Indian languages



Confidence Thresholds

Assign confidence scores and flag uncertain results



Privacy-by-Design

Minimize data storage and follow security best practices



IMPLEMENTATION ROADMAP

Phased development for sustainable impact

Phase 1
(MVP)

- Text + Image Verification
- Hindi + Marathi + English

01

Phase 2
(Expansion)

- Video Context
- Scam Detection

02

Phase 3
(Scale)

- Browser Extension
- More Indian Languages

03



FROM CHALLENGE TO IMPACT

A reliable verification process

Challenge

Misinformation
& UncertaintyAI Verification
LayerAnalyze • Cross-check
Verify with evidenceEvidence-backed
ResultTrusted, accurate and
clear information

EXPECTED OUTCOME

- ✓ Technically feasible
- ✓ Scalable and cost-effective
- ✓ Mitigated risks
- ✓ High potential social impact
- ✓ A more informed and safer India

05

IMPACT & BENEFITS

PEOPLE | PROGRESS | A MORE INFORMED INDIA



TARGET AUDIENCE

For a safer and more informed society



Citizens



Students



Rural &
Semi-Urban
Users



Journalists
& Media
Professionals



Fact-Checkers
& Researchers



Educators



Government
Teams



POTENTIAL IMPACT

Creating a more truthful digital ecosystem



Safer Digital Information

Reduce misleading content and viral misinformation



Regional Language Access

Verification beyond English (Indian languages)



Better Digital Literacy

Help users understand why content is questionable



Faster Verification

Evidence and trusted sources in one place



Public Information Safety

Verify government schemes, jobs, health and disaster-related updates



KEY BENEFITS

Positive change across multiple dimensions



SOCIAL BENEFITS

Builds digital trust,
responsible sharing
and a more informed
society.



ECONOMIC BENEFITS

Helps identify financial
scams, fake jobs and
fraudulent offers,
saving money and
resources.



EDUCATIONAL BENEFITS

Supports students,
teachers and media
literacy, encouraging
critical thinking.



PUBLIC SAFETY BENEFITS

Helps verify health,
disaster and emergency
information, contributing
to safer communities.



IMPACT CHAIN

Turning verification into real-world impact



DETECT

Identify questionable
content



UNDERSTAND

Analyse and explain
the context



VERIFY

Cross-check with
trusted sources



CORRECT

Provide facts and
clear information



PREVENT SPREAD

Stop misinformation
from spreading

RESEARCH & REFERENCES

IEEE

- Deep Learning for Fake News Detection — IEEE Access (2024)
- → AI/Transformer-based misinformation detection.
- Dual Attention Fusion Networks — IEEE ISCC (2021)
- → Multimodal text + image fake-news detection.
- Explainable Multimodal Fake News Detection — IEEE Access (2025)
- → Explainable and transparent AI verification.

ScienceDirect

- Multilingual Misinformation Detection — Applied Soft Computing (2026)
- → Multilingual & low-resource language challenges.
- Multimodal Fake News Detection — Results in Engineering (2025)
- → Text, image, video & audio verification.
- Explainable AI for Fake News Detection — EAAI (2023)
- → Explainable and trustworthy predictions.

